ACADEMIC TASK-1

(Case-Based Assignment)

Of CSE 320

Title name

COMPUTER SCIENCE AND ENGINEERING

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1. Introduction

1.1: Purpose:

This document provides a detailed description of the requirements for the Online Shoping Cart System using cloud Microservices.

The Purpose of this system is to develop a Scalable, secure, and high-Performance e-Commerce platform that allows customers to Purchase Products online using a cloud-based distributed architecture.

This SRS is intended for:

→ Developers
→ Project managers
→ System architects
→ Testers.
→ Academic evaluators.

1.2 Scope of The System:

The System will:

→ allow Customers to create accounts and log-in securely
→ Allow browsing and seaching Products.

→ Allow adding Products to shoping cart

→ Allow placing orders and making payments

→ Allow order tracking.

→ Allow administrators to mange Products and users

→ Send notification via email/SMS

The System will be implemented using cloud microservices architecture, where each major functionality will be developed as an independent Service.

1.3 objectives :—

The main objectives are :

→ High scalability using cloud infrastructure

→ independent deployment of services.

→ Secure Payment Processing

→ fast response time

→ High availability (24/7).

→ fault Tolerance.

2. Overall Description :

2.1 System Architecture Overview :—

The system will follow a Microservice Overview deployed in the cloud.

Each Service will :

→ Have its own database

→ Communicate using REST APIs

→ Be independently scalable

→ Be Containerized using Docker (optional)

Architecture Diagram : →

Monolithic architecture

User Interface

Business Logic

Data Access Layer

DB

→ Handles payment Success / failure

2.2 Major Microservices :—

① User Service :—

→ Handles user registration and login
→ Manages authentication and authorization
→ stores user Profile information.

② Products Service :—

→ Manages Products Catalog
→ stores Product details (Price, image, stock)
→ Handles search and filtering

③ Cart Service :—

→ Manages shoping cart items
→ Updates total price
→ Calculates total Price.

④ Order Service :—

→ Handles order placement
→ Generate order ID
→ Maintains order History

⑤ Payment Service :—

→ Integrates with Payment gateway

→ Handles payment Success / failure

→ update order status.

⑥ Notification Service :—

→ Sends email Confirmation

→ Sends SMS updates

2.3 Users Classes and characteristic :—

Customer :

→ Registers and logs in

→ Browses Products.

→ adds item to cart

→ Makes payments

→ Tracks orders.

Admin :

→ Adds / updates / deletes Products

→ Manages users

→ Monitors orders.

Delivery Staff :

→ Views assigned orders

→ Updates delivery status.

→ System is

A [illegible] :—

2.4 Operating Environment :—

→ Frontend : Web Browsers (Chrome, Edge, Firefox)

→ Backend : Cloud Server (AWS / Azure / GCP)

→ Database : MySQL / MongoDB

→ Communication : REST APIs.

→ Security : HTTPS.

3. Functional Requirements :

3.1 User Registration :—

→ System shall allow new users to register.

→ System shall validate email and password.

→ System shall store encrypted passwords.

3.2 Login Authentication :—

→ System shall display all Products

→ System shall allow filtering by category

→ System shall allow searching by Product name.

3.3 Product Browsing :—

→ System shall verify credentials
→ System shall generate authentication token (JWT)
→ System shall prevent unauthorized access.

3.4 Shopping Cart Management :—

→ Add Product to cart
→ Remove Product from cart
→ Update product quantity
→ Calculate total cost automatically

3.5 Order Processing :—

→ User can confirmed order
→ System generates unique order ID.
→ Order status → Paid → Shipped → Delivered.

3.6 Payment Processing :—

→ Redirect to secure payment gateway
→ Verify transaction
→ update order status after Payment.

→ Backup database daily

3.7 Notification Handling :—

→ Send confirmation email after order.
→ Send SMS for delivery updates.

4. Non-Functional Requirements :

4.1 Performance :—

→ Response time < 3 sec
→ Support at least 1000 concurrent users.

4.2 Security :—

→ HTTPS encryption
→ Secure token authentication (JWT)
→ Secure payment gateway integration.
→ Role-based access control.

4.3 Scalability :—

→ Services must scale independently
→ Load balancer support required
→ Auto-scalling in cloud.